

02302N - Operating Systems Security (3 Cr; 2 L, 2 P): Introduction to Operating Systems Security, File system security, Access control mechanisms, memory protections, Inter-process communication vulnerabilities, User and kernel mode securities, Hardware mechanisms, Virtualization, Mobile operating system security. Prerequisite(s): 00307N, 02201N

02302N Operating Systems Security

Faculty of Computer & Data Sciences
Alexandria National University
Spring 2025

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Policy:

- Lecture:
 - Monday 9:00 -10:40 (Group 1)
 - Monday 11:00 -12:40 (Group 2)
- Practice: TA (Eng. Rawan El Gendy)
 - Thursday 9:00 -10:40 (Group 1)
 - Thursday 11:00 -12:40 (Group 2)
- Text: This lecture notes
- Grading: (3 Credit hour)
 - (20 Marks) Attendance (both lectures & practices) + Assignments
 - (20 Marks) Midterm Exam
 - (60 Marks) Final Exam
- Course Web page:
Ms Team: **02302N – Operating Systems Security S25**

join code

hnnyy7w

You should attend in
your group **only**;
otherwise you will be

absent!!!

Content

- Introduction to Operating Systems Security
- Processes & Memory Protections
- File System Security
- Access Control
- Inter-process communication vulnerabilities
- Virtualization
- Mobile Operating Systems

References

- *Operating Systems: Three Easy Pieces*, Remzi H. Arpaci-Dusseau and Andrea C. Arpaci-Dusseau, Arpaci-Dusseau Books; October, 2023 (Version 1.10); <http://www.ostep.org>
- *Computer and Information Security Handbook-Third Edition*; Edited by Jhon R. Vacca; 2017 Elsevier Inc.
- Security in Computing 6th edition; Charles P. Pfleeger, Shari Lawrence Pfleeger, Lizzie Coles-Kemp; © 2024 Pearson Education, Inc.
- Computer Security and the Internet: Tools and Jewels from Malware to Bitcoin (2nd Edition), Paul van Oorschot, Springer, 2021.
- Stallings, William. Operating systems : internals and design principles. 7th ed. Printice Hall 2012
- Operating System Security; Trent Jaeger; www.morganclaypool.com, 2008 ebook; A Publication in the Morgan & Claypool Publishers series SYNTHESIS LECTURES ON INFORMATION SECURITY, PRIVACY AND TRUST.
- On the Evolution of Buffer Overflows, Matthias Vallentin; May 20, 2007https://matthias.vallentin.net/course-work/buffer_overflows.pdf

Operating Systems Security

- What is Security?Review
 - Security is a state of well-being of information and infrastructures in which the possibility of a threat or successful yet undetected attack on information and services is kept low or tolerable.
 - Principle of Adequate Protection
 - “Security is economics”: Don’t spend \$100,000 to protect a system that can only cause \$1,000 in damage
 - Security rests on *confidentiality, authenticity, integrity, and availability* (CIA). i.e. Protect:
 - **Confidentiality** is the concealment of information or resources.
 - **Authenticity** is the identification and assurance of the origin of information.
 - **Integrity** refers to the trustworthiness of data or resources in terms of preventing improper and unauthorized changes.
 - **Availability** refers to the ability to use the information or resource desired.

Security: Some Terminology

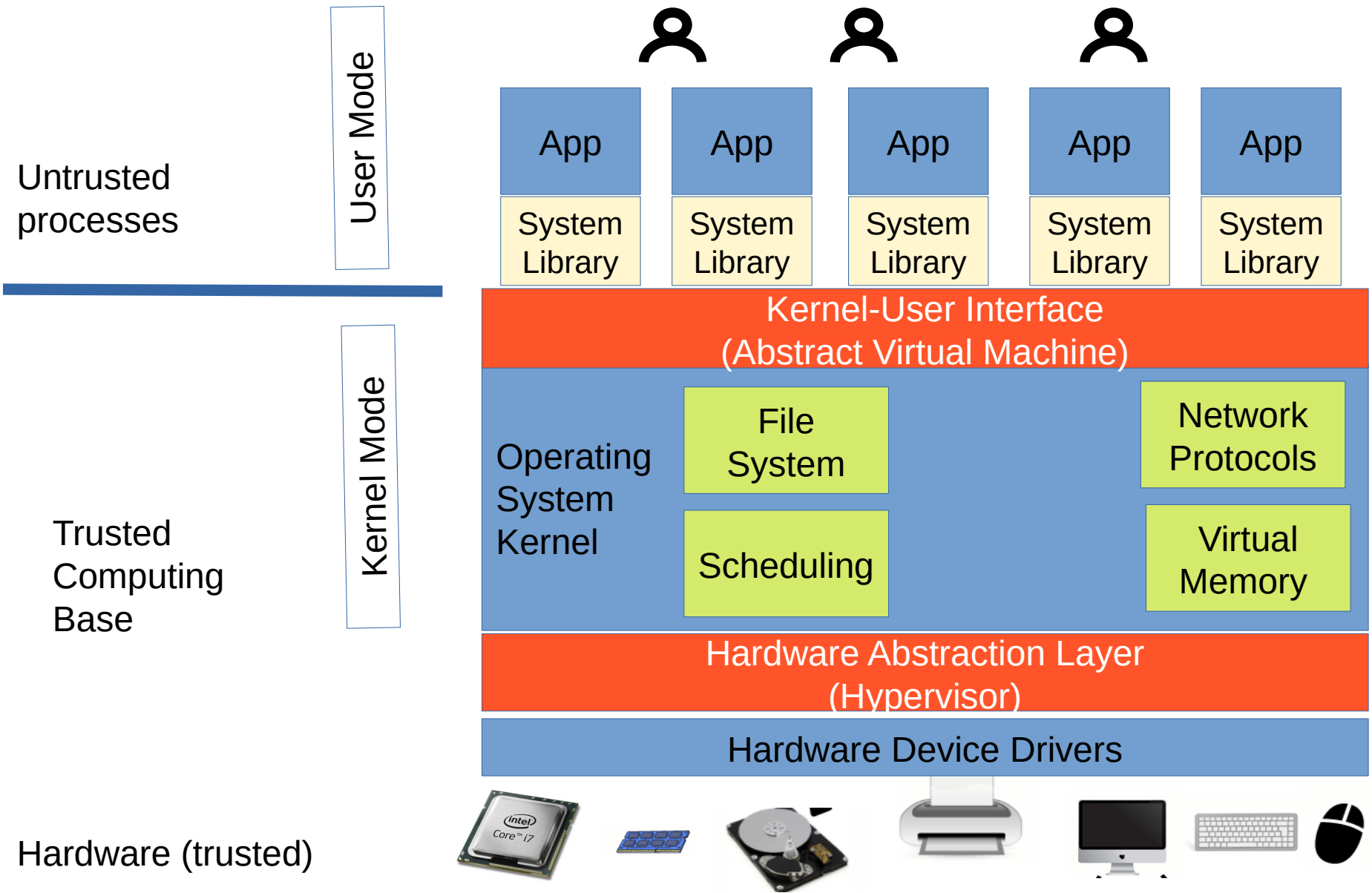
- A **threat** is a potential violation of security.
 - Usually due to flaws in design, implementation, and/or operation.
 - There are four major categories of threats: Interception, Interruption, Modification, Fabrication.
- **Threat model**: When designing security of a system, we need to state the threat model
 - Set of threats to defend against (Whom do we want to prevent from doing what?)
- **Adversaries**: Those who are trying to mess with us.
 - Amateurs, fired x-employee, crackers, organized crime, government “cyber-warriors”, terrorists.
- An **attack** is any action that violates security.
 - An attack has an implicit concept of “intent”. For example, Router miss-configuration or server crash can cause loss of availability, they are threats but are not attacks.

Security: Some Terminology

- **Assets**: Things we might want to protect, such as hardware, software, data.
- **Vulnerabilities**: Weaknesses in a system that may be able to be exploited in order to cause loss or harm.
 - Example: a web server that doesn't authenticate its users
- **Exploit**: A piece of code that makes use of an existing vulnerability.

Operating Systems

- What is OS? Review
 - Controller & manager of all resources (CPU, Memory, Storage, I/O, Network, ...).
 - These resources are shared between all programs running on the system, including the OS itself.
 - Provide interface for users & programs.(UI, API)
- Modern OS is designed as modular & layered.
 - Each module exists in one layer and performs a well-defined operation.
 - Each layer has a privilege level with its modules provide specific services to the modules in upper layers.



Operating Systems Security

- Since OS is the “master” of the machine; it is a primary target of attacks. For Secure operation, OS must have the properties:
 - Protecting & defending against all sorts of miss behavior.
 - Protect itself in order to protect its users and resources.
- OS is the 1st line of defense, if the OS is compromised, attackers can :
 - impersonate users,
 - destroy/corrupt/disclose sensitive information,
 - bypass security measures of the OS,
 - participate in DDoS attack or send spam over networks.

Defense in Information Systems

- For security requirements, we want to protect all of our assets: Hardware, software, data. Many ways to do this using different kinds of controls:
 - *Physical*: e.g. off-site backups
 - *Policies and Procedures*: e.g. rules about choosing passwords
 - *Hardware*: e.g. fingerprint readers and smart tokens
 - *Software*: e.g. operating systems separate users' actions from each other
 - *Cryptography*: e.g. protecting data by making it unreadable to an attacker
- In fact, we usually use more than one of these controls (***defense in depth***)
- In this course, we are interested software controls specially in the operating systems.

“Trusted” OS or “Secure” OS

- Security professionals “prefer” to speak of ***trusted*** instead of ***secure*** operating systems.
 - The word “secure” reflects a dichotomy: Something is either secure or not secure - nothing in between.
 - A trusted system refers to one that meets the intended security requirements (the cost level of security incidence).
- In other words, trust is perceived by the system’s user, not by its manufacturer.
- So, OS must be **Trusted Computing Base** (TCB) with the following requirements:
 - **Complete mediation:** A TCB must mediate all access to the underlying protected resources. All requests for access to resources must pass through the operating system.
 - **Tamper-proof:** If untrusted code executing on top of the OS can tamper with the OS then we can no longer trust the OS.
 - **Correctness:** If we are going to rely on the OS to ensure that protected resources get used in a proper way, it is imperative that this access control is implemented correctly.

Trusted Computing Base (TCB)

Complete Mediation



Authentication,
Authorization and
Access Control.

Tamper proof



Isolation between
OS process and
user processes.

Correctness



Isolation between
user processes.

Isolation between processes usually accomplished by protecting memory address space and files of each process from illegal access.

Special hardware elements are needed to help OS isolate processes.